

Finite-Dimensional Quotients of Separable-Fibre Continuous-Trace Algebras Have BCP

Run fa_banach_001, lane 19

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Source Problem

The source paper is Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257. After proving that a commutative unital C^* -algebra $C(K)$ has BCP/UBCP exactly when K has a countable π -basis, the authors ask for a noncommutative version:

$\{U_\alpha\}_{\alpha \in A}$ of K is called a π -basis (weak basis) for K if every nonempty open subset of K contains some U_α in $\mathcal{U}$ [37, Page 15].

In the setting of commutative C^* -algebra, [25, Theorem 2.1] can be reformulated as follows. Note that [25, Theorem 2.1] only deals with real continuous functions. The complex case is a direct consequence of the real case.

Corollary 3.9. *Let $\mathcal{A}$ be a unital commutative C^* -algebra. Then the followings are equivalent:*

- (1) $\mathcal{A}$ has the BCP;
- (2) the pure state space $\mathfrak{P}(\mathcal{A})$ has a countable π -basis;
- (3) $\mathcal{A}$ has the UBCP.

In particular, there exist commutative C^* -algebras containing no minimal projections but having the BCP. For example, the uniform norm closure of the linear span of Haar functions [24, p.150].

We conclude this section by the following question.

Question 3. *It will be interesting to find a non-commutative version of Corollary 3.9, that is, how to characterize those C^* -algebras $\mathcal{A}$ having BCP/UBCP.*

Definitions

Definition 1. *A C^* -algebra A has the ball-covering property, or BCP, if the unit sphere of A is covered by countably many open balls none of which contains the origin. It has the uniform ball-covering property, or UBCP, if the balls can be chosen with a common positive gap from the origin: for centers c_n and radii r_n ,*

$$\inf_n (\|c_n\| - r_n) > 0.$$

Definition 2. *Let J be a closed ideal of a C^* -algebra A . We say that J is γ -norming in A , where $0 < \gamma \leq 1$, if*

$$\sup\{\|\pi(a)\| : \pi \in \widehat{J}\} \geq \gamma\|a\| \quad (a \in A).$$

Statement

This packet removes the norming hypothesis from the previous finite-quotient packet. The result is still a partial answer to the original source problem, because the ideal is assumed continuous trace and the quotient is finite dimensional.

Theorem 1 (finite-dimensional quotient continuous-trace criterion). *Let*

$$0 \longrightarrow I \longrightarrow A \xrightarrow{q} F \longrightarrow 0$$

be an exact sequence of C^ -algebras. Assume that F is finite dimensional and that I is continuous trace with elementary fibres $\mathcal{K}(H_x)$, where each H_x is separable and nonzero.*

If $\text{Prim}(I)$ has a countable π -basis, then A has UBCP. Conversely, if A has BCP, then $\text{Prim}(I)$ has a countable π -basis. Consequently, in this finite-dimensional quotient continuous-trace class,

$$A \text{ has BCP} \iff A \text{ has UBCP} \iff \text{Prim}(I) \text{ has a countable } \pi\text{-basis}.$$

Idea of the Proof

The earlier matrix-block ballast theorem proves the result under the extra assumption that I is uniformly norming in A . At first the missing case seems to require a separate quotient-point cover: perhaps some elements are almost invisible on $\text{Prim}(I)$ and are normed mainly by the finite quotient.

The key observation is that this is not a new covering problem. The invisible part is exactly the annihilator

$$J = \text{Ann}_A(I) = \{a \in A : aI = Ia = 0\}.$$

The Busby map $\tau : F \rightarrow Q(I) = M(I)/I$ has kernel $q(J)$. Thus, after splitting off the finite-dimensional ideal J and replacing I by

$$I^\# = I \oplus J,$$

the remaining quotient has injective Busby map. Injective C^* -homomorphisms are isometric, so the action on the ideal sees the quotient norm. This makes $I^\#$ uniformly norming in A , and the previous norming theorem applies.

Imported Norming Theorem

We use the following result from the packet `2311.14257_norming_finite_quotient_bcp`.

Proposition 1 (norming finite-dimensional quotient criterion). *Let*

$$0 \longrightarrow B \longrightarrow E \longrightarrow G \longrightarrow 0$$

be exact. Assume that G is finite dimensional, B is continuous trace with separable nonzero elementary fibres, $\text{Prim}(B)$ has a countable π -basis, and B is γ -norming in E for some $\gamma > 0$. Then E has UBCP.

The proof of Proposition 1 is the matrix-block ballast argument: choose a bounded linear section of the finite quotient, approximate the compact ideal component locally, and insert a rank-one spike aligned with the local quotient-lifted operator. The proof does not require a $*$ -splitting of the quotient.

Proof

We first isolate the exact obstruction to norming.

Lemma 1 (Busby kernel equals the quotient of the annihilator). *Let*

$$0 \longrightarrow I \longrightarrow A \xrightarrow{q} F \longrightarrow 0$$

be exact, and let $\phi : A \rightarrow M(I)$ be the canonical action homomorphism. Put

$$J = \text{Ann}_A(I) = \{a \in A : aI = Ia = 0\}.$$

Let $\tau : F \rightarrow Q(I) = M(I)/I$ be the Busby map. Then

$$\ker \phi = J, \quad \ker \tau = q(J).$$

In particular, if F is finite dimensional, then J is finite dimensional.

Proof. By definition, $\phi(a)$ is the multiplier of I obtained by left and right multiplication by a . Hence $\phi(a) = 0$ exactly when $aI = Ia = 0$, so $\ker \phi = J$.

If $j \in J$, then $\phi(j) = 0$, and therefore

$$\tau(q(j)) = \phi(j) + I = 0.$$

Thus $q(J) \subseteq \ker \tau$.

Conversely, suppose $\tau(q(a)) = 0$. Then $\phi(a)$ belongs to the copy of I inside $M(I)$. Hence there is $i \in I$ such that $\phi(a) = \phi(i)$. Therefore $a - i \in \ker \phi = J$, and

$$q(a) = q(a - i) \in q(J).$$

So $\ker \tau = q(J)$.

Finally $J \cap I = \{0\}$: if $x \in J \cap I$, then $xx^* = 0$ because $x^* \in I$ and $xI = 0$, hence $x = 0$. Thus $q|_J$ is injective. Since $q(J) \subset F$, finite dimensionality of F implies finite dimensionality of J . $\square$

Lemma 2 (finite annihilator split). *In the setting of Lemma 1, assume F is finite dimensional. Then $J = \text{Ann}_A(I)$ is a finite-dimensional unital ideal. If $p \in J$ is its identity, then p is central in A , and*

$$A = J \oplus A_0, \quad A_0 = \{a \in A : pa = 0\}.$$

Moreover $I \subset A_0$, and $A_0/I \cong F/q(J)$.

Proof. A finite-dimensional C^* -algebra is unital; let p be the unit of J . For any $a \in A$, both pa and ap belong to the ideal J . Since p is the identity on J ,

$$pa = (pa)p = p(ap) = ap.$$

Thus p is central in A .

Every $a \in A$ decomposes as

$$a = pa + (a - pa),$$

with $pa \in J$ and $p(a - pa) = 0$. The intersection $J \cap A_0$ is zero because p acts as the identity on J . Hence $A = J \oplus A_0$ as a C^* -direct sum. Since J annihilates I , we have $pI = 0$, so $I \subset A_0$.

The map $A_0 \rightarrow F/q(J)$, $a_0 \mapsto q(a_0) + q(J)$, is onto because

$$q(a) = q(a - pa) + q(pa), \quad q(pa) \in q(J).$$

Its kernel is I : if $a_0 \in A_0$ and $q(a_0) \in q(J)$, then $a_0 - j \in I$ for some $j \in J$. Applying p gives

$$0 - j = p(a_0 - j) \in I,$$

so $j \in I \cap J = \{0\}$, and $a_0 \in I$. □

Lemma 3 (the enlarged ideal is norming). *Assume the hypotheses of Theorem 1. Let*

$$J = \text{Ann}_A(I), \quad I^\# = I \oplus J.$$

Then $I^\#$ is a continuous-trace ideal of A with separable nonzero elementary fibres. If $\text{Prim}(I)$ has a countable π -basis, then so does $\text{Prim}(I^\#)$. Moreover $I^\#$ is 1-norming in A .

Proof. By Lemma 2, $A = J \oplus A_0$, $I \subset A_0$, and

$$0 \longrightarrow I \longrightarrow A_0 \longrightarrow F/q(J) \longrightarrow 0$$

is exact. The Busby map of this reduced extension is the quotient of τ by its kernel $q(J)$, hence is injective by Lemma 1. Since it is an injective C^* -homomorphism, it is isometric.

Let $a_0 \in A_0$, and put

$$y = a_0 + I \in A_0/I \cong F/q(J).$$

Let $\phi_0 : A_0 \rightarrow M(I)$ be the canonical action. The norm of $\phi_0(a_0)$ is the supremum of the norms of a_0 in the primitive representations over $\text{Prim}(I)$. Write this number as

$$\alpha(a_0) = \sup\{\|\pi(a_0)\| : \pi \in \widehat{I}\}.$$

The Busby value satisfies

$$\|y\| = \|\tau_0(y)\| = \|\phi_0(a_0) + I\|_{Q(I)} \leq \|\phi_0(a_0)\| = \alpha(a_0).$$

The norm in A_0 is the supremum over primitive representations that either restrict nontrivially to I , or factor through A_0/I . Therefore

$$\|a_0\| = \max\{\alpha(a_0), \|y\|\} = \alpha(a_0).$$

Thus I is already 1-norming in A_0 .

Now take an arbitrary $a \in A$, and write $a = j + a_0$ with $j \in J$ and $a_0 \in A_0$. Because $A = J \oplus A_0$ is a C^* -direct sum,

$$\|a\| = \max\{\|j\|, \|a_0\|\}.$$

The primitive representations of $I^\# = J \oplus I$ see the two summands, so

$$\begin{aligned} \sup_{\rho \in \widehat{I^\#}} \|\rho(a)\| &= \max \left\{ \|j\|, \sup_{\pi \in \widehat{I}} \|\pi(a_0)\| \right\} \\ &= \max\{\|j\|, \|a_0\|\} = \|a\|. \end{aligned}$$

Hence $I^\#$ is 1-norming in A .

Finally, $I^\# = I \oplus J$ is a direct sum of the given continuous-trace algebra and a finite-dimensional C^* -algebra, hence is continuous trace. Its elementary fibres are those of I together with finitely many matrix algebras, all over separable nonzero Hilbert spaces. Also

$$\text{Prim}(I^\#) = \text{Prim}(I) \sqcup \text{Prim}(J)$$

as a topological disjoint union, and $\text{Prim}(J)$ is finite discrete. Thus a countable π -basis for $\text{Prim}(I)$, together with the finitely many singleton points of $\text{Prim}(J)$, gives a countable π -basis for $\text{Prim}(I^\#)$. $\square$

Proof of Theorem 1. Assume first that $\text{Prim}(I)$ has a countable π -basis. By Lemma 3, the enlarged ideal

$$I^\# = I \oplus \text{Ann}_A(I)$$

is continuous trace with separable nonzero elementary fibres, has a countable π -basis on its primitive spectrum, and is 1-norming in A . The quotient

$$A/I^\# \cong F/q(\text{Ann}_A(I))$$

is finite dimensional. Proposition 1, applied to

$$0 \longrightarrow I^\# \longrightarrow A \longrightarrow A/I^\# \longrightarrow 0,$$

therefore gives UBCP for A .

Conversely, if A has BCP, the primitive π -base obstruction for C^* -algebras from the earlier lane 19 obstruction packet gives a countable π -basis for $\text{Prim}(A)$. Since $\text{Prim}(I)$ is an open subspace of $\text{Prim}(A)$, it inherits a countable π -basis. This proves the converse implication. The equivalence follows because UBCP implies BCP. $\square$

Scope and Limitations

This is a genuine removal of the norming hypothesis for finite-dimensional quotients over the positive continuous-trace/separable-fibre class. It is not claimed as a full solution to the source paper's broad noncommutative problem: the proof still uses continuous trace structure and a finite-dimensional quotient.

The next natural targets are:

1. finite composition series whose subquotients are of the above form;
2. Fell or type-I algebras with separable irreducible representations;
3. a general obstruction theorem showing when non-type-I primitive fibres force failure of BCP.

Novelty and Search Bounds

This packet was produced as a strengthening of the earlier lane 19 norming finite-quotient packet. During the surrounding run, the local indexes were searched for arXiv:2311.14257, finite-dimensional quotients, finite-dimensional extensions, unitization, matrix-block ballast, continuous trace, Busby maps, annihilator ideals, BCP, UBCP, and primitive π -bases. Earlier web searches on June 17, 2026 for BCP finite codimensional permanence, finite-dimensional C^* -quotient splitting, and Fell/type-I local continuous-trace structure found the source paper, vector-valued

$C_0(K, X)$ BCP stability arXiv:2111.04921, Calkin/operator-space BCP papers arXiv:2412.07137 and arXiv:2507.02261, and Fell-algebra structure arXiv:1207.2540, but no exact finite-dimensional quotient continuous-trace BCP theorem. No claim of complete novelty is made without human literature review.

Human Review Focus

The proof reduces the new theorem to one standard extension-theoretic calculation. The most important points to check are:

1. the equality $\ker \tau = q(\text{Ann}_A(I))$;
2. centrality of the unit of the finite-dimensional annihilator ideal;
3. the assertion that the reduced Busby map is injective and therefore isometric;
4. the norm formula $\|a_0\| = \max\{\alpha(a_0), \|a_0 + I\|\}$ for the reduced extension.

If these pass review, the finite-dimensional quotient obstruction from the previous packet is removed.

References

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- [3] Run `fa_banach_001`, lane 19, *Unitizations of Separable-Fibre Continuous-Trace Algebras Have BCP*, solution packet `2311.14257_unitization_continuous_trace_bcp`, 2026.
- [4] Run `fa_banach_001`, lane 19, *Primitive π -Base Obstruction for C^* -Algebras with BCP*, solution packet `2311.14257_primitive_pi_base_obstruction`, 2026.
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